Deriving the Approximate Adiabatic Condition

1 Setup

The quantum state evolves according to the Schrödinger equation

$$i\frac{d}{dt}|\psi(t)\rangle = H(t)|\psi(t)\rangle.$$

Let the instantaneous eigenstates and eigenvalues satisfy

$$H(t)|E_n(t)\rangle = E_n(t)|E_n(t)\rangle, \qquad \langle E_m(t)|E_n(t)\rangle = \delta_{mn}.$$

Expand the state in the instantaneous eigenbasis:

$$|\psi(t)\rangle = \sum_n c_n(t)e^{-i\theta_n(t)}|E_n(t)\rangle, \qquad \theta_n(t) = \int_0^t E_n(\tau)\,d\tau.$$

Substituting into the Schrödinger equation and projecting onto $\langle E_m(t)|$ gives

$$\dot{c}_m(t) = -\sum_n c_n(t)\langle E_m(t)|\dot{E}_n(t)\rangle e^{i\int_0^t [E_m(\tau)-E_n(\tau)]\,d\tau}.$$

2 First-Order Transition Amplitude

Assume the system starts in its ground state:

$$c_0(0) = 1, \qquad c_{m\neq 0}(0) = 0.$$

To first order, approximate $c_0 \approx 1$ and neglect excited-state contributions. For $m \neq 0$,

$$\dot{c}_m(t) \approx -\langle E_m(t)|\dot{E}_0(t)\rangle e^{i\int_0^t [E_m(\tau)-E_0(\tau)]\,d\tau}.$$

Integrating,

$$c_m(t) \approx -\int_0^t f(\tau)e^{i\Phi(\tau)}\,d\tau,$$

where

$$f(t) = \langle E_m(t)|\dot{E}_0(t)\rangle, \qquad \Phi(t) = \int_0^t g(\tau)\,d\tau, \qquad g(t) = E_m(t) - E_0(t).$$

3 Integration by Parts

Since

$$\frac{d}{dt}e^{i\Phi(t)} = ig(t)e^{i\Phi(t)},$$

we can write

$$c_m(t) \approx -\int_0^t \frac{f(\tau)}{ig(\tau)}\frac{d}{d\tau}e^{i\Phi(\tau)}\,d\tau.$$

Integration by parts yields

$$\begin{aligned} c_m(t) \approx & -\left[\frac{f(\tau)}{ig(\tau)}e^{i\Phi(\tau)}\right]_0^t \\ & + \int_0^t \frac{d}{d\tau}\left(\frac{f(\tau)}{ig(\tau)}\right)e^{i\Phi(\tau)}\,d\tau. \end{aligned}$$

The exponential oscillates at a rate determined by the energy gap. For a sufficiently large gap and sufficiently smooth evolution, these oscillations can suppress the integral through cancellation. The integral is not automatically negligible; its size requires appropriate smoothness and gap assumptions.

This motivates the heuristic estimate

$$|c_m| \sim \frac{|f|}{|g|}.$$

4 Why the Gap Is Squared

Differentiate the eigenvalue equation

$$H|E_0\rangle = E_0|E_0\rangle$$

and project onto $\langle E_m|$, with $m \neq 0$. Using orthogonality,

$$\langle E_m|\dot{H}|E_0\rangle + E_m\langle E_m|\dot{E}_0\rangle = E_0\langle E_m|\dot{E}_0\rangle.$$

Therefore,

$$\boxed{\langle E_m|\dot{E}_0\rangle = \frac{\langle E_m|\dot{H}|E_0\rangle}{E_0 - E_m}}$$

Consequently,

$$|f| = \frac{|\langle E_m|\dot{H}|E_0\rangle|}{|g|}.$$

Substituting into the transition-amplitude estimate gives

$$\boxed{|c_m| \sim \frac{|\langle E_m|\dot{H}|E_0\rangle|}{|E_m - E_0|^2}}$$

Thus, the approximate adiabatic condition is

$$\boxed{\frac{|\langle E_m|\dot{H}|E_0\rangle|}{|E_m - E_0|^2} \ll 1, \qquad m \neq 0.}$$

One factor of the inverse gap arises from the changing eigenstates; the other arises from suppression due to phase oscillations.

Caveat. This is a heuristic derivation, not a rigorous adiabatic theorem. Rigorous guarantees require bounds on the gap, the Hamiltonian's derivatives, and the evolution time.